

Supporting Information

Stable Hydrogen Evolution from CdS Modified CuGaSe₂ Photoelectrode under Visible Light Irradiation

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Preparation of CuGaSe₂ films on Mo coated soda lime glass plates by co-evaporation method

Elemental Cu, Ga and Se were evaporated using conventional effusion cells and deposition rate of these metals were determined using thickness monitor before the sample preparation. During the deposition of CGSe, a deposition rate of Cu, Ga and Se were kept at 0.03, 0.06, and 1.0 nm/s. CGSe thin films were deposited at 723 K for 10 min followed by deposition at 823 K for 80 min.

Chemical Bath Deposition (CBD) of CdS

The solution containing distilled water, 0.375 M SC(NH₂)₂ (Kanto, 98 %), 0.025 M Cd(CH₃COO)₂ (Kanto, 98 %) and 14 wt% ammonia were used as a chemical bath. CdS deposition was performed at 348 K for 1 – 10 min followed by calcination in air at 573 K for an hour.

Electrode preparation for PEC measurements

A lead wire was connected to prepared electrode using indium followed by covering the unnecessary part of the electrode with epoxy resin before PEC measurements. The electrodes were exposed to the light from 300 W Xe lamp in the solution containing 10 mM H₂PtCl₆ (Kanto, 98.5%), and 0.1 M Na₂SO₄ (Wako, 99%) at the potential of -0.56 V vs. Ag/AgCl until the saturation of photocurrent. No particles were found in SEM images of prepared electrode surfaces however XPS found existence of Pt. Judging from SEM images and XPS results, the particle size of deposited Pt on prepared electrode is smaller than 10 nm.

Photoelectrochemical measurements

PEC measurements were conducted using a normal three-electrode system cell. Prepared photoelectrode, Pt wire, and Ag/AgCl electrode were used as a working, counter, and reference electrode. pH adjusted 0.1 M Na₂SO₄ aqueous solution by adding NaOH was employed as an electrolyte. Solar simulator or 150 W Xe lamp or 300 W Xe lamp equipped with filters (HOYA, L42 and OPTO-LINE, CM1) were employed as a light source. For current-potential (*I-E*) curve measurement, the potential of specimen was swept to positive direction with a rate of 5 mVs⁻¹ under intermittent light irradiation with a period of 6 s. 150 W Xe lamp was used for durability test.

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Calculation of band alignment at electrode/electrolyte interface

Band alignment at electrode/liquid interfaces were calculated using Poisson's equation which given as

$$\frac{d^2\varphi}{dx^2} = - \frac{eN_D}{\varepsilon\varepsilon_0} \quad (S1)$$

for CdS layers and

$$\frac{d^2\varphi}{dx^2} = \frac{eN_A}{\varepsilon\varepsilon_0} \quad (S2)$$

for CGSe layers where φ and x of potential and position. Concentrations of donor in CdS and acceptor in CGSe was supposed to be $N_A = N_D = 10^{16} / cm^{-3}$. Dielectric constant ε was assumed to be 10. Results of calculations are shown in Figure 4.

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Results of Sample Analysis

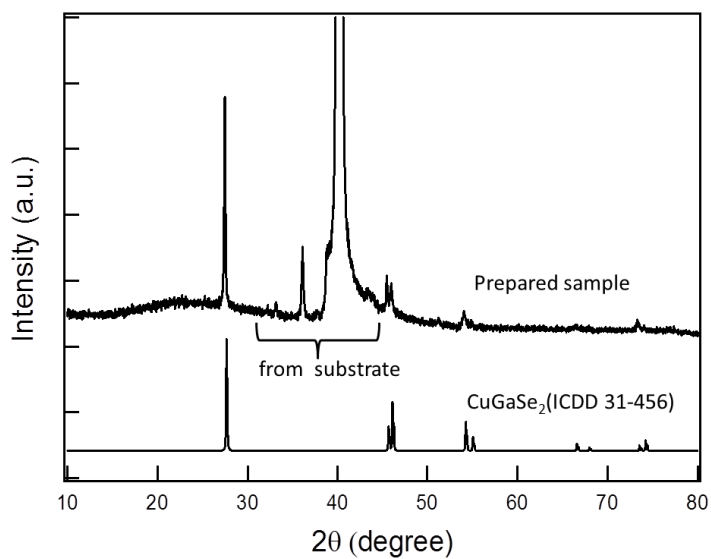


Figure S1. XRD pattern from prepared CuGaSe₂ film on Mo coated soda lime glass plate. Cu Kα1 line was used.

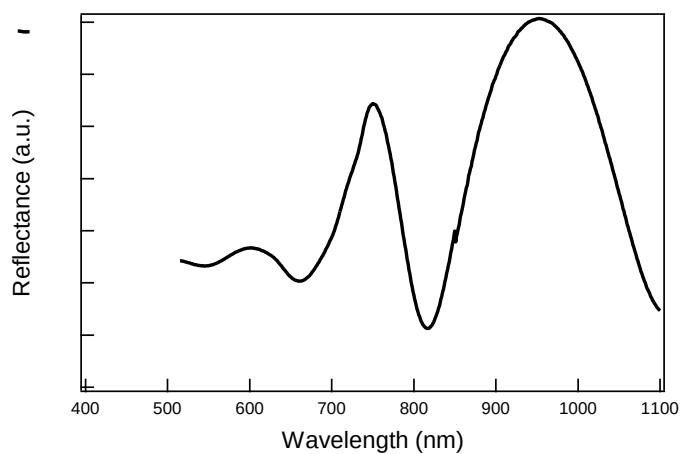


Figure S2. UV-vis. DRS of prepared CuGaSe₂ specimen. Judging from the damping of interference, bandgap value was decided to be 1.7 eV.

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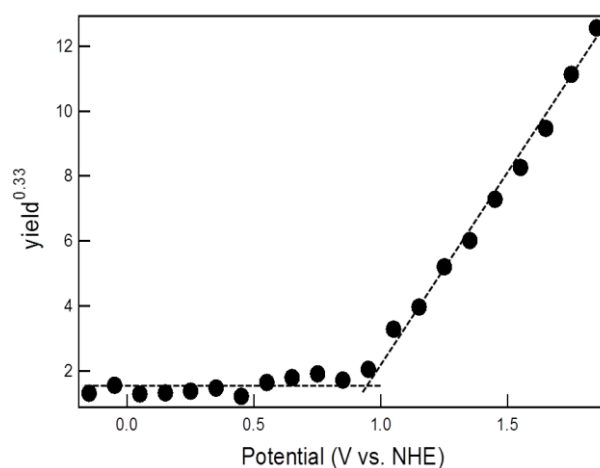


Figure S3. PESA spectrum of prepared CuGaSe₂ film. The potential of valence band maximum was estimated to be 0.9 V vs. NHE.

I-E curve under simulated sunlight

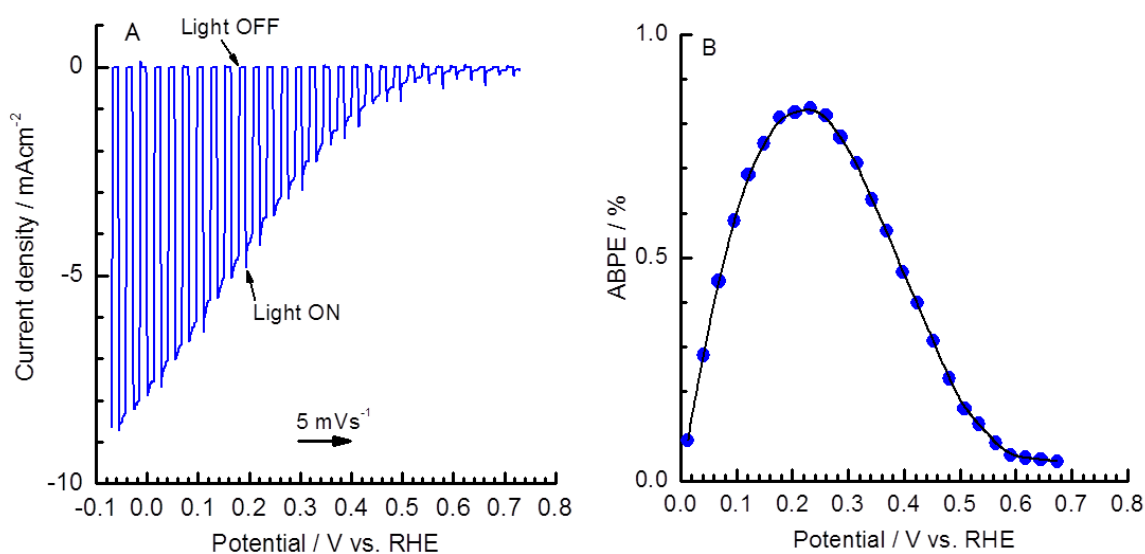


Figure S4. *I-E* curve (A) and ABPE (b) at actual potential of Pt/CdS/CGSe electrode with a CBD time of 1 min under AM1.5G irradiation. 0.1 M Na₂SO₄ aqueous solution with pH adjustment to 9 by addition of NaOH was used as an electrolyte. AM1.5G light was turned on and off with a period of 6 s. The electrode potential was swept in the positive direction at a rate of 5 mV s⁻¹.

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Cross Sectional SEM-EDS mapping

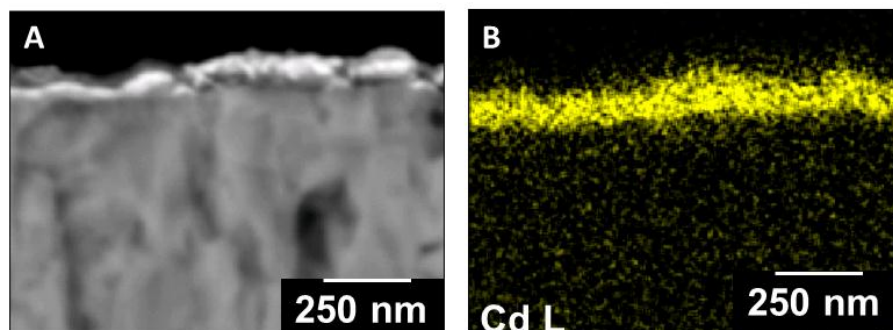


Figure S5. A cross sectional SEM image (A) and a result of EDX mapping (B) of CdS/CGSe sample with CBD for 1 min. A 40 nm thick CdS layers were found.

Mott-Schottky plot of prepared CuGaSe₂ electrode

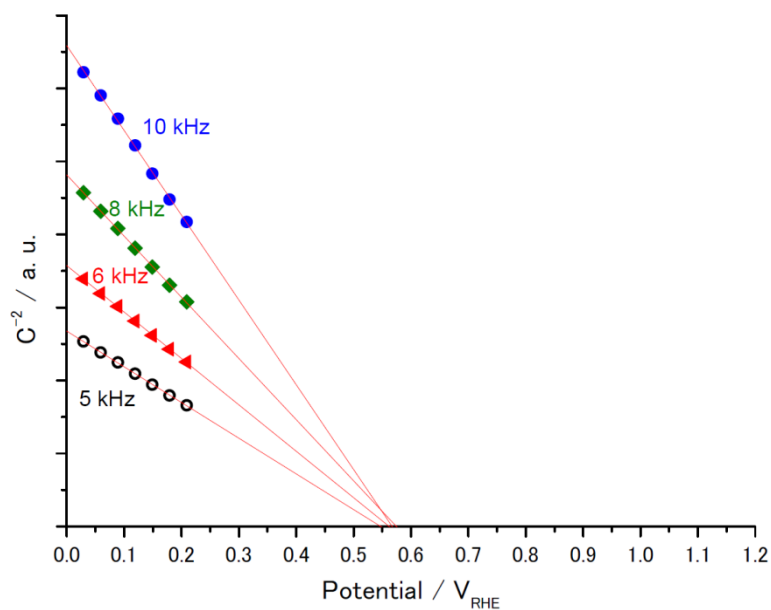


Figure S6. Mott-Schottky plot of prepared CuGaSe₂ electrode. 0.1 M Na₂SO₄ aqueous solution with pH adjustment to 9 by addition of NaOH was used as an electrolyte. A flat band potential was estimated to be 0.55 V vs. RHE. An acceptor concentration was estimated to be in the order of 10^{17} cm^{-3} from the slope value of C^{-2} .

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XPS and SEM results before and after the durability test

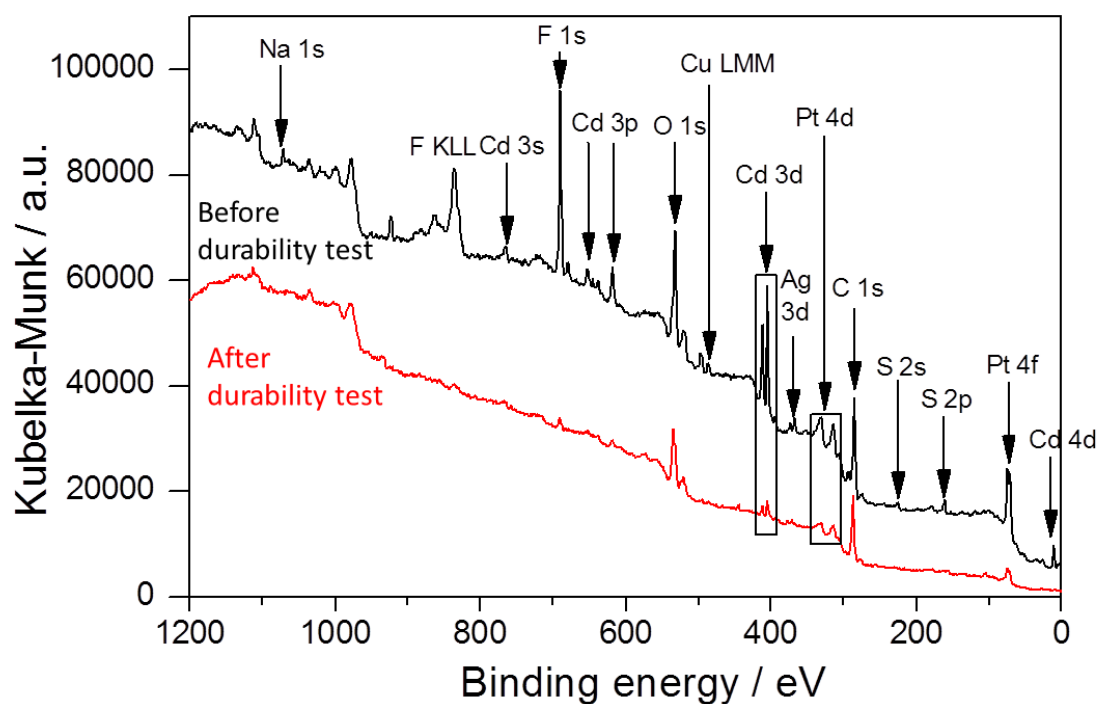


Figure S7. XPS measurement results of Pt/CdS/CGSe sample before and after durability test. Signals from Ga were not found while signals from Cd and Pt were found. It should be noted that signals from S became weaker after durability test.

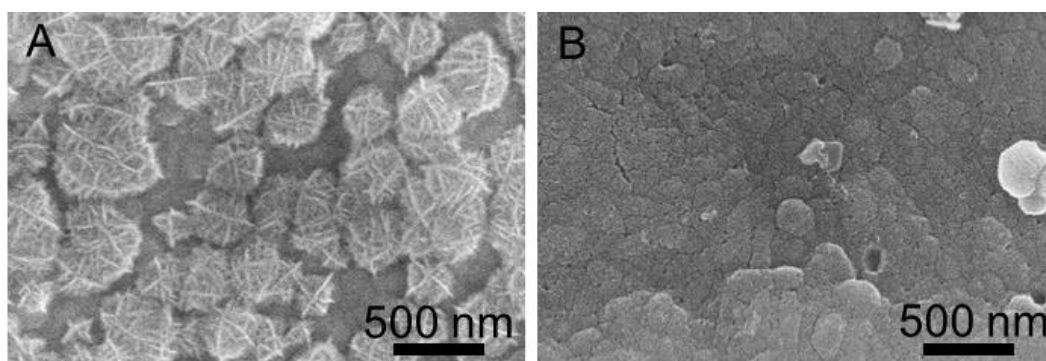


Figure S8. SEM image of Pt/CdS/CGSe electrode before (A) and after (B) durability test.

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Results of Gas product analysis

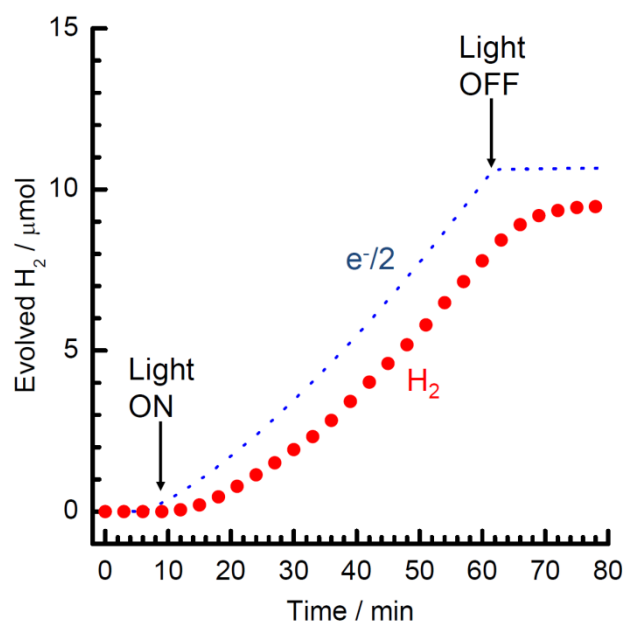


Figure S9. Amount of evolved H₂ from Pt/CdS/CuGaSe₂ electrode (0.09 cm²). The electrolyte was a 0.1 M aqueous Na₂SO₄ solution with pH adjusted to 9.5 by NaOH addition. A 300 W Xe lamp equipped with a cut-off filter (HOYA, L42) was used. The electrode potential was set to -0.1 V_{RHE} during the measurement. Faradaic efficiency was ~90% in total.